

PHYSICS OF THE CARDIOVASCULAR SYSTEM

Part 1

The cells of the body act like individual engines. In order for them to function they must have: -

1. Fuel from our food to supply energy.
2. O₂ from the air we breathe to combine with the food to release energy.
3. A way to dispose of the by-products of the combustion (mostly CO₂, H₂O, and heat).

Since the body has many billions of cells an elaborate transportation system is needed to deliver the fuel and O₂ to the cells and remove the by-products. The blood performs this important body function. Blood represents about 7% of the body mass or about 4.5kg (~4.4 liters) in a 64kg person. The blood, blood vessels, and heart make up the cardiovascular system (CVS).

Major components of the cardiovascular system

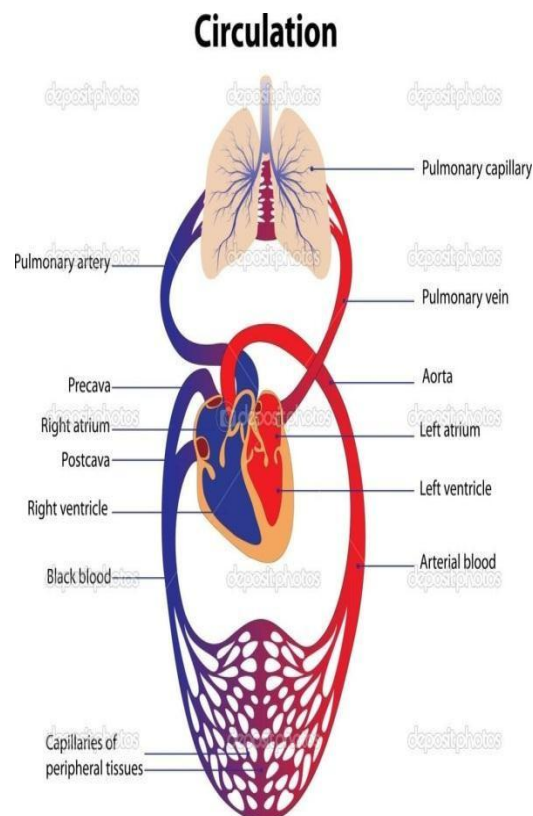
The heart is basically a double pump; it provides the force needed to circulate the blood through the two major circulatory systems: -

1. The pulmonary circulation in the lungs.
2. The systemic circulation in the rest of the body.

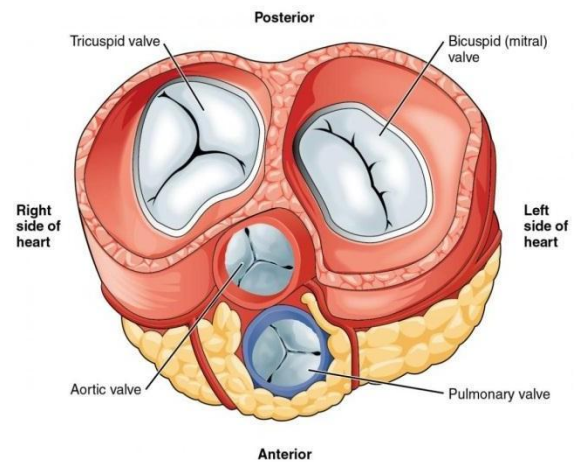
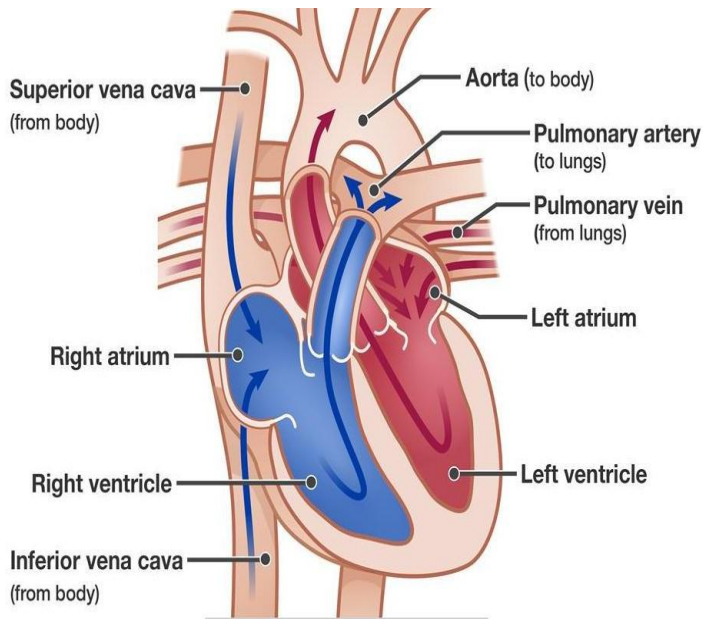
The blood in a normal individual circulates through one system before being pumped by the other section of the heart to the second system.

Let us start with the blood in the left side of the heart and follow its circulation through one complete loop.

- **The blood is pumped by the contraction of the heart muscles from the left ventricle at a pressure of about 125 mm Hg into a system of arteries that subdivided into smaller and smaller arteries (arterioles) and finally into a very fine meshwork of vessels called the capillary bed.**
- **During the few seconds it is in the capillary bed the blood supplies O₂ to the cells and picks up CO₂ from the cells.**
- **After passing through the capillary bed the blood collects in small veins (venules) that gradually combine into larger and larger veins before it enters the right side of the heart via two main veins—the superior vena cava and the inferior vena cava.**
- **The returning blood is momentarily stored in the reservoir (the right atrium). And during a weak contraction (5-6 mm Hg) the blood flows into the right ventricle.**
- **On the next ventricular contraction this blood is pumped at a pressure of about 25 mm Hg via the pulmonary arteries to the capillary system in the lungs, where it receives more O₂ and where some of the CO₂ diffuses into the air in the lungs to be exhaled.**
- **The freshly oxygenated blood then travels via the main veins from the lungs into the left reservoir of the heart (left atrium); during the weak atrial contraction (7-8 mmHg) the blood flows into the left ventricle.**
- **On the next ventricular contraction this blood is again pumped from the left side of the heart into the general circulation.**



The heart has a system of valves that, if functioning properly, permit the blood to flow only in the correct direction. If these valves become diseased and do not open or close properly the pumping of the blood becomes inefficient.



The blood volume is not uniformly divided between the pulmonary and systemic circulation, but it is: -

Blood Circulation 100/100					
Systemic Circulation 80/100			Pulmonary Circulation 20/100		
Arteries 15/100	Capillaries 10/100	Veins 75/100	Arteries 46.5/100	Capillaries 7/100	Veins 46.5/100

Q) Calculate the mass of blood in all circulation of a person his body mass is 80kg.

Mass of Blood					
$80 \times 7/100 = 5.6\text{kg}$					
Mass of Blood in Systemic Circulation			Mass of Blood in Pulmonary Circulation		
$5.6 \times 80/100 = 4.48\text{kg}$			$5.6 \times 20/100 = 1.12\text{kg}$		
Arteries	Capillaries	Veins	Arteries	Capillaries	Veins
$4.48 \times 15/100$ $= 0.672\text{kg}$	$4.48 \times 10/100$ $= 0.448\text{kg}$	$4.48 \times 75/100$ $= 3.360\text{kg}$	$1.12 \times 46.5/100$ $= 0.521\text{kg}$	$1.12 \times 7/100$ $= 0.078\text{kg}$	$1.12 \times 46.5/100$ $= 0.521\text{kg}$

Q) The mass of the pulmonary blood of a person is 1.5kg, find: -

- 1) The mass of this person.
- 2) The mass of his systemic blood.

$$\text{Pulmonary Mass} = \text{Blood Mass} \times 20/100$$

$$1.5 = \text{Blood Mass} \times 20/100$$

$$\text{Blood Mass} = 7.5\text{kg}$$

- 1) The mass of this person: -

$$\text{Blood Mass} = \text{Body Mass} \times 7/100$$

$$7.5 = \text{Body Mass} \times 7/100$$

$$\text{Body Mass} = 107\text{kg}$$

- 2) The mass of his systemic blood: -

$$\text{Systemic Mass} = \text{Blood Mass} \times 80/100$$

$$\text{Systemic Mass} = 7.5 \times 80/100 = \text{Systemic Mass} = 6\text{kg}$$

To the eye blood appears to be a red liquid slightly thicker than water. When examined by various physical techniques it is found to consist of several different components. The red color is caused by the red blood cells (erythrocytes). A nearly clear fluid called blood plasma accounts for the other 55%. The combination of red blood cells and plasma causes blood to have flow properties different from those of a fluid like water.

Beside red blood cells and plasma, there are some important blood components, such as the white blood cells (leukocytes), present in small amounts. The blood also contains platelets. Platelets are involved in the clotting function of blood.

O₂ and CO₂ exchange in the capillary system

Oxygen and carbon dioxide also diffuse through tissue. The most probable distance D that a molecule will travel after N collisions with other molecules with an average distance λ between collisions is: -

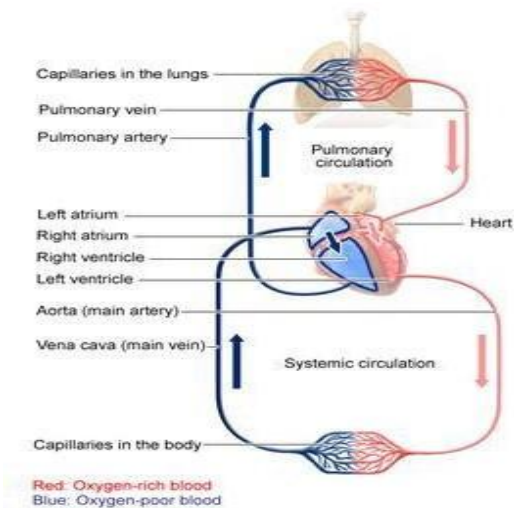
$$D = \lambda \sqrt{N}$$

In tissue the density of the molecules is about 1000 times greater than in air; therefore, λ is much longer in air than in tissue.

A typical value for λ in water, which can serve as a model for tissue, is about 10^{-11} m, and a molecule makes about 10^{12} collisions/sec. Thus after 1sec in water the most probable diffusion distance is about 10^{-5} m or about a factor of 10^3 less than in air.

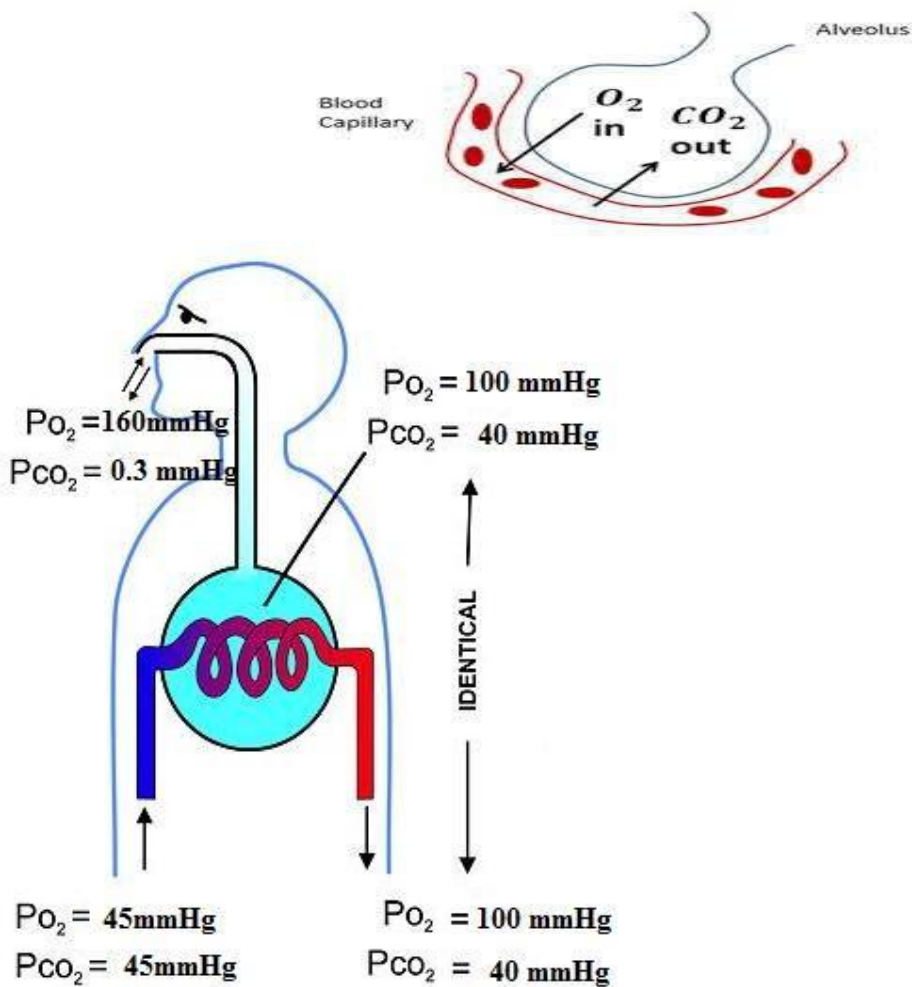
$$\begin{aligned} D &= \lambda \sqrt{N} \\ &= 10^{-11} \sqrt{10^{12}} = 10^{-11} \times 10^6 = 10^{-5} \text{ m} \end{aligned}$$

This very short diffusion distance is the primary reason that the capillaries in tissue must be very close together.



O₂ and CO₂ exchange in the capillary system

In respiration, oxygen (O_2) is required to enter cells, while waste carbon dioxide (CO_2) must be excreted;



Oxygen and carbon dioxide move independently of each other; they diffuse down their own pressure gradients.

During each inhalation, at rest, approximately 500 ml of fresh air flows in through the nose.

The alveolar partial pressure of oxygen remains very close to (100 mmHg), and the partial pressure of carbon dioxide varies minimally around (40 mmHg) throughout the breathing cycle (of inhalation and exhalation).

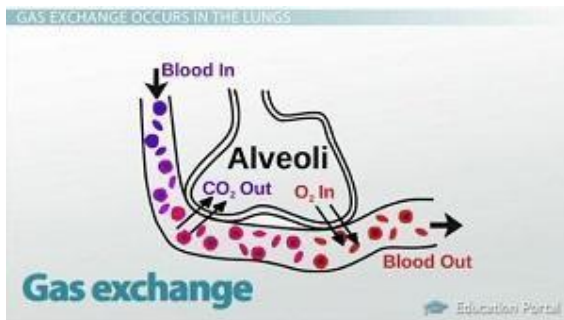
As blood leaves the lungs through the pulmonary veins, the venous $PO_2=100\text{mmHg}$, whereas the venous $PCO_2=40\text{mmHg}$.

As blood enters the systemic capillaries, the blood will lose oxygen and gain carbon dioxide because of the pressure difference between the tissues and blood.

In systemic capillaries, $PO_2=100\text{mmHg}$, but in the tissue cells, $PO_2=40\text{mmHg}$. This pressure gradient drives the diffusion of oxygen out of the capillaries and into the tissue cells. At the same time, blood

$PCO_2=40\text{mmHg}$ and systemic tissue $PCO_2=45\text{mmHg}$.

The pressure gradient drives CO_2 out of tissue cells and into the capillaries. The blood returning to the lungs through the pulmonary arteries has a venous $PO_2=40\text{mmHg}$ and a $PCO_2=45\text{mmHg}$. The blood enters the lung capillaries where the process of exchanging gases between the capillaries and alveoli begins again .

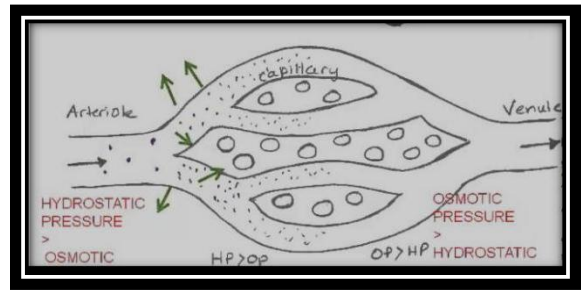


Not all capillaries are carrying blood at any one time. In resting muscle only 2 to 5% of the capillaries are functional.

Starling's law of capillary describes the flow of fluids into and out of the capillaries.

Fluid movement through the capillary wall is the result of two pressures: -

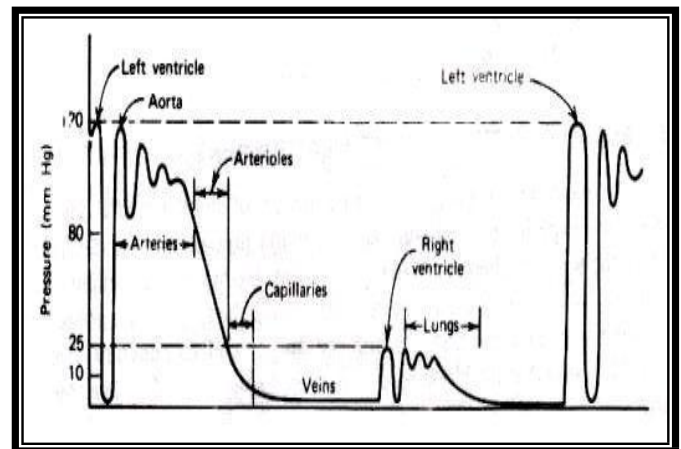
1. The hydrostatic pressure p across the capillary wall forcing fluids out of the capillary.
2. The osmotic pressure π bringing fluids in.



The heart

In a typical adult each contraction of the heart muscles forces about 80ml (about one-third of a cup) of blood through the lungs from the right ventricle and a similar volume to the systemic circulation from the left ventricle. In the process the heart does work.

The pressures in the two pumps of the heart are not the same. In the pulmonary system the pressure is quite low because of the low resistance of the blood vessels in the lungs.



The maximum pressure (systole), typically about 25mmHg, is about one-fifth of that in the systemic circulation. In order to circulate the blood through the much larger systemic network the left side of the heart must produce pressures that are typically about 120mmHg at the peak (systole) of each cardiac cycle. During the resting phase (diastole) of the cardiac cycle the pressure is typically about 80mmHg.

The muscle driving the left ventricle is about three times thicker than that of the right ventricle. In addition, the circular shape of the left ventricle is more efficient for producing high pressure than the elliptical shape of the right ventricle.

The work done W by a pump working at a constant pressure P is equal to the product of the pressure and the volume pumped ΔV , or $W=P\Delta V$. We can estimate the physical work done by the heart by multiplying its average pressure by the volume of blood that is pumped.

Actually the pumping action takes place in less than one-third of the cardiac cycle and the heart muscle rests for over two-thirds of the cycle.

Q) The heart rate of a person is 120 pulse/min; calculate the action time and the resting time of heart muscle.

$$120\text{pulse}/1\text{min} = 120\text{pulse}/60\text{sec}$$

$$2\text{pulse}/1\text{sec} = 1\text{pulse}/0.5\text{sec}$$

$$1\text{pulse} = 1/3\text{contraction} + 2/3\text{relaxation}$$

$$0.5 \times 1/3 = 0.17\text{sec (the time of contraction)}$$

$$0.5 \times 2/3 = 0.33\text{sec (the time of relaxation)}$$

Q) Person has a systolic pressure 150mmHg, diastolic pressure 100mmHg, heart rate 90/min. Calculate the work done and the efficiency of the lower left half of the heart if the energy consume is 6 Watt.

$$\text{Work done} = P\Delta V$$

$$P \text{ average} = (\text{systolic} + \text{diastolic})/2$$

$$P \text{ average} = (150 + 100)/2$$

$$P \text{ average} = 125 \text{ mmHg}$$

$$1 \text{ mmHg} = 1330 \text{ dyne/cm}^2$$

$$125 \text{ mmHg} = 166250 \text{ dyne/cm}^2$$

$$\Delta V = \text{amount of blood in each beat/sec} \times \text{number of beat/sec}$$

$$\Delta V = 80 \text{ ml/beat/sec} \times (90 \text{ beats/min}) / (60 \text{ sec/min})$$

$$\Delta V = 120 \text{ cm}^3$$

$$W = P\Delta V$$

$$\text{Work} = 166250 \text{ dyne/cm}^2 \times 120 \text{ cm}^3$$

$$\text{Work} = 19950000 \text{ dyne.cm}$$

$$1 \text{ erg} = 1 \text{ dyne.cm}$$

$$\text{Work} = 19950000 \text{ erg}$$

$$1 \text{ erg} = 10^{-7} \text{ Joule}$$

$$\text{Work} = 1.995 \text{ Joule}$$

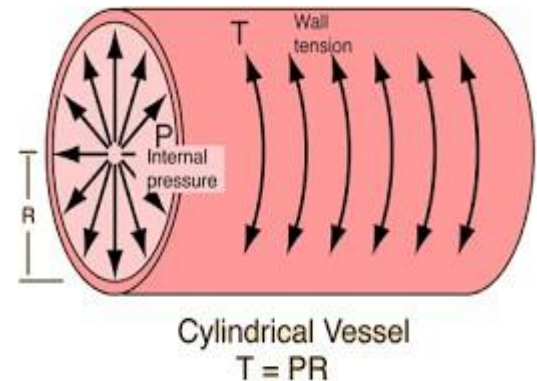
$$\text{Efficiency} = (\text{Work done} / \text{Energy consume}) \times 100\%$$

$$\text{Efficiency} = (1.995/6) \times 100\% = 33.25 \%$$

Pressure across the blood vessel wall (Transmural Pressure)

The greatest pressure drop in the cardiovascular system occurs in the region of the arterioles and capillaries.

In order to understand why they do not burst we must discuss the law of Laplace, which tells us how the tension in the wall of a tube is related to the radius of the tube and the pressure inside the tube.



Consider a long tube of radius R carrying blood at pressure P . We can calculate the tension T in the wall ($T=RP$).

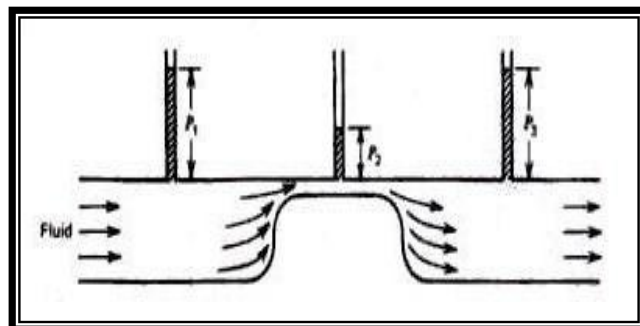
Bernoulli's principle applied to the cardiovascular system

Whenever there is a rapid flow of a fluid such as air or water, the pressure is reduced at the edge of the rapidly moving fluid.

Bernoulli's principle is based on the law of conservation of energy. Pressure in a fluid is a form of potential energy PE since it has the ability to perform useful work. In a moving fluid there is kinetic energy KE due to the motion. This kinetic energy can be expressed as energy per unit volume such as ergs per cubic centimeter.

If fluid is flowing through the frictionless tube, the velocity increases in the narrow section and the increased kinetic energy KE of the fluid is obtained by a reduction of the potential energy PE of the pressure in the tube.

As the velocity reduces again on the far side of the restriction the kinetic energy is converted back into potential energy and the pressure increases again as indicated on the manometers.



We can calculate the average K.E per unit volume of (1 g) ~ (1 cm³) of blood as it leaves the heart .remember that:

$$K.E = \frac{1}{2} mv^2$$

Since the average velocity is about 30 cm/sec

$$K.E = \frac{1}{2} \times 1 \times (30)^2 = 450 \text{ ergs}$$

This K.E is equivalent to a P.E of 450 dyne/cm².

During heavy exercise the velocity of blood being pumped by the heart may be five times its average value during rest.

Notice that the blood velocity is related in an inverse way to the total cross-sectional area of the vessels carrying the blood. The velocity equals the flow rate divided by the cross- sectional area. The average velocity in the aorta is about 30 cm/sec; that in a capillary is only about 1 mm/sec. It is in the capillaries that the exchange of O₂ and CO₂ takes place, and this low velocity allows time for diffusion of the gases to occur.

9- How fast does your blood flow?

As the blood moves from the heart, the arteries branch and re-branch many times to carry out blood to the various tissues. The smallest blood vessels are the capillaries. The figure below shows that the blood velocity is related in an inverse way to the total cross- sectional area of the vessels carrying the blood

Velocity= Flow rate / cross-section area

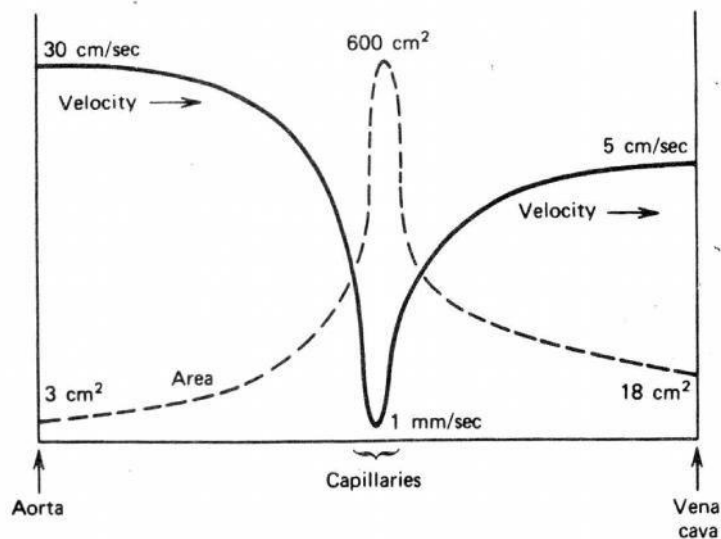


Figure 4 . The dashed curve shows schematically the change in cross-sectional area of the circulatory system. The velocity of blood flow (solid line) decreases as the total cross-sectional area increases. The total cross-sectional area is obtained by adding the areas of all the vessels at a given distance from the heart. Note that the vena cava returning the blood to the heart has a much larger cross-sectional area than the aorta.

In order to understand the laws that control the flow of blood in the circulatory system we must study the Poiseuille's law.

11- Poiseuille's Law:

Poiseuille's law states that the flow through a given tube depends on the pressure difference from one end to the other ($P_A - P_B$), the length of the tube (vessel), the radius (R) of the tube (vessel) and viscosity η of the fluid. If the pressure difference is doubled the flow rate also doubled. The flow varies inversely with the length and viscosity if either is doubled; the flow rate is reduced by one -half. The flow rate increased as the radius of the tube increased if the radius is doubled the flow rate increases, by 2^4 or a factor of 16.

$$\text{Flow rate} = (P_A - P_B) \times \frac{\pi}{8} \times \frac{1}{\eta} \times \frac{R^4}{L}$$

Poiseuille's Law applies to rigid tubes of constant radius since:-

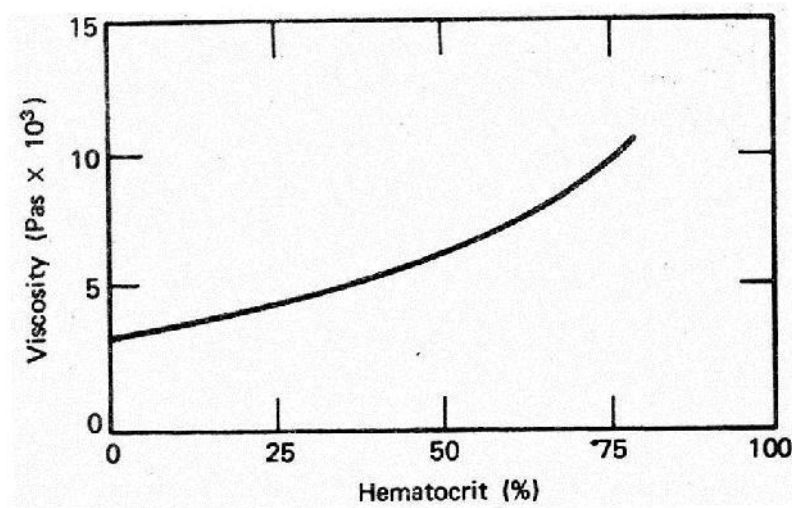
- 1 The major arteries have elastic walls and expand slightly at each heart beat; blood flow in the circulatory system does not obey the law exactly.
- 2 The blood viscosity changes slightly with flow rate, however this effect is negligible.

-In SI units the flow rate will be in m/sec if $P_A - P_B$ is in N/m^2 , η is in pas, and R and L in meters.

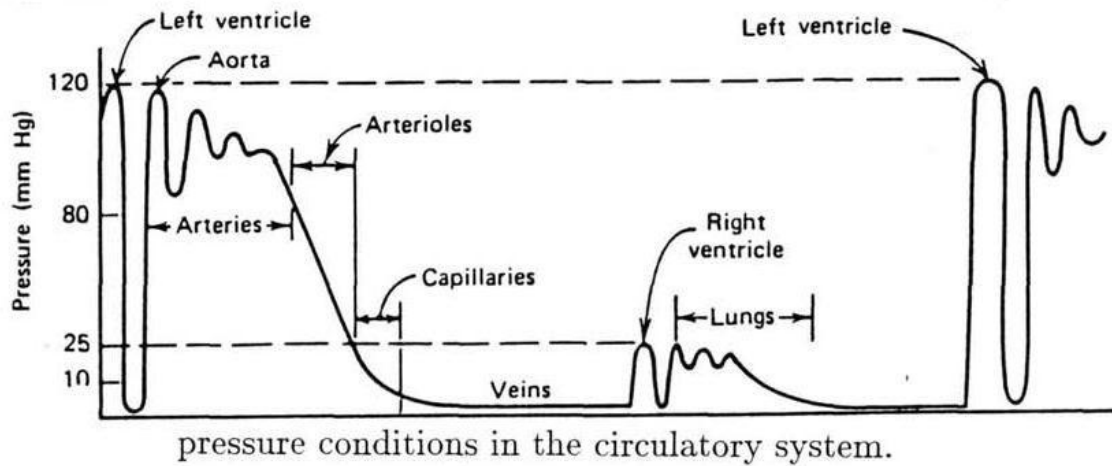
Average velocity in the aorta is about 30 cm/sec. That is in capillaries it is only about 1 mm/sec. It is in the capillaries that the exchange of O_2 and CO_2 takes place and this low velocity allows times for diffusion of the gases to occur.

Blood viscosity:

The cgs unit used to measure viscosity is the *poise*. The SI unit for viscosity is the pascal second (pas), which equals 10 poises. The viscosity of blood is typical 3×10^{-3} to 4×10^{-3} pas but depends on the percentage of RBCs in the blood (the hematocrit). As the hematocrit increases, the viscosity increase. Show the figure below.



Persons with the disease *polycythemia vera* in which there is an overproduction of RBCs have a high hematocrit and often have circulatory problems. The viscosity of the blood also depends on temperature. As blood gets colder, the viscosity increases and this further reduce the blood supply to cold hands and feet. A change from $37^\circ C$ to $0^\circ C$ increases the viscosity of blood by a factor of 2.5. In addition to viscosity, other factors affect the flow of blood in the vessels: the pressure difference from one end to the other, the length of the vessel, and its radius.



The work W done by a pump working at a constant pressure P is equal to the product of pressure and volume pumped ΔV or

$$W = P \Delta V$$

$$\text{Average } P =$$

$$10\text{cmHg } \Delta V =$$

$$80\text{ml}$$

$$W = \rho g h \Delta V$$

$$W = 13.6 \text{ g/cm}^3 \times 980 \text{ cm/s}^2 \times 10\text{cm} \times 80\text{cm}^3$$

$$= 1.1 \times 10^7 \text{ ergs}$$

$$\text{Or } = 1.1 \text{ J/s or a}$$

power of 1.1W

Pressure across the blood vessel wall

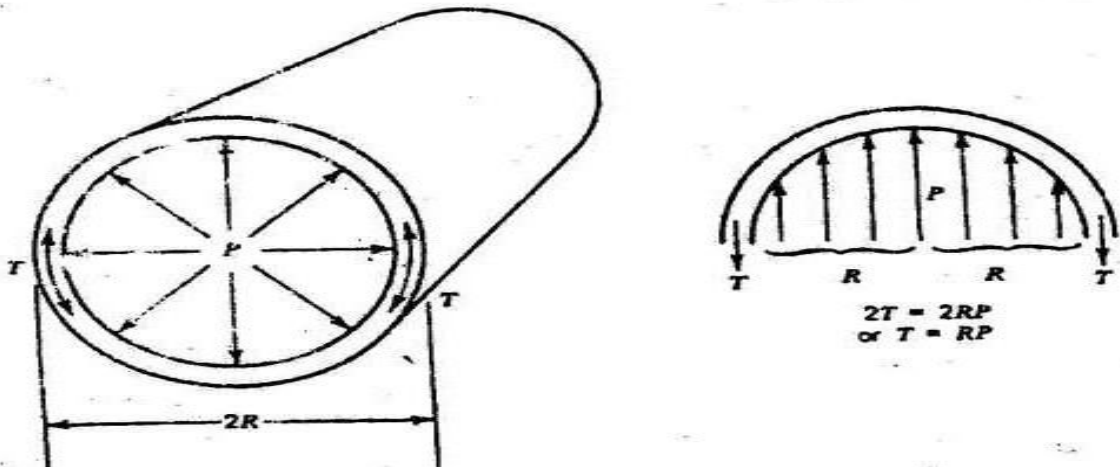
The greatest pressure drop in the cardiovascular system occurs in the region of the arterioles and capillaries. The capillaries have very thin wall that permit easy diffusion of O₂ and CO₂. In order to understand why they do not burst, we must discuss wall of a tube is related to the radius of the tube and the pressure inside the tube

$$T = RP$$

Consider along tube of radius R carrying blood at pressure P. We calculate the tension T in the wall. We can divide the tube in half as shown in figure. The force per unit length pushing upward is 2RP. There is a tension force T per unit length at each edge that holds the top half of the tube to bottom half. Since the wall is in equilibrium the force pushing the two halves a part is equal to the tension forces holding them to gather or

$$2T = 2RP$$

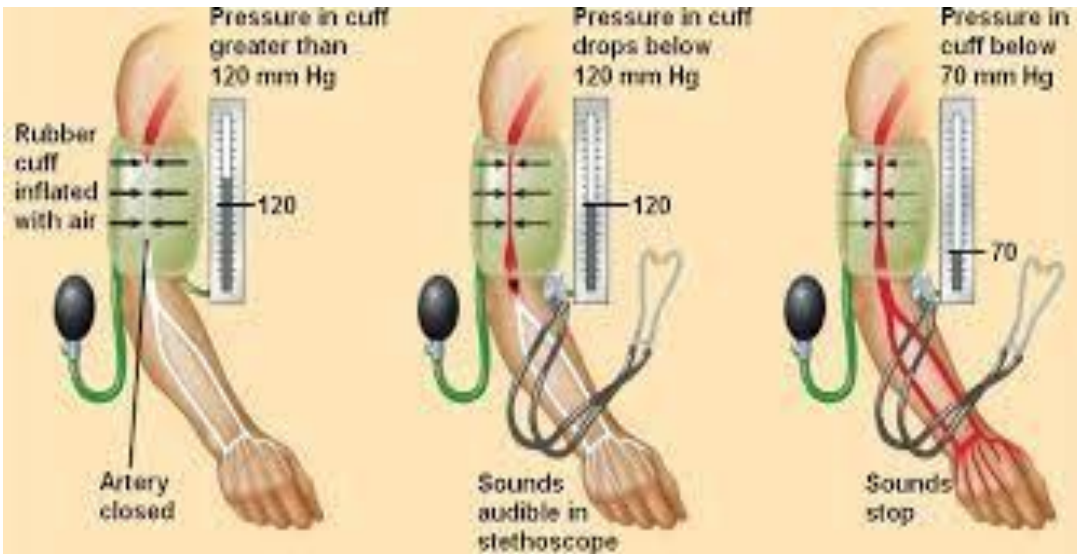
or $T = RP$ Laplace equation



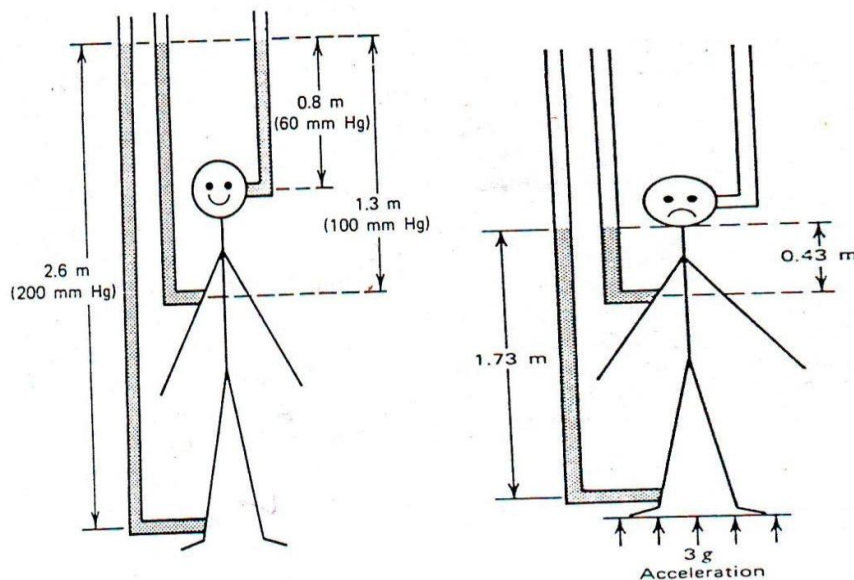
Blood pressure and its measurement

The blood pressure is measured by sphygmomanometer.

1. The arterial blood flow to the arm is blocked by an inflated cuff.
2. As the air gradually released, the stethoscope placed over the brachial artery is used to listen for korotkoff sounds.
3. The pressure at which the sound appear and change are noted on the gauge.



The pressure varies from one point to another because of gravitational forces
 The greater pressure P in the foot is due to the gravitational force ($\rho g h$) produced by the column of blood (of height h) between the heart and foot added to the pressure at the heart.,
 similarly, the decreased pressure in the head is due to the elevation of the head over the heart



If gravity on earth suddenly became three times greater, blood would rise only about 4.3cm above the heart and it would not reach the brain of a standing person.
 This situation can be produced artificially by acceleration the body at 3g in vertical direction. This condition produces pooling of blood in the legs.